VISHNU KUMAR

Software Engineer – Python | Backend & API Development

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PROFESSIONAL SUMMARY

Final-year B.Tech Computer Science student and Python software engineer with hands-on experience designing, coding, and deploying REST APIs, back-end services, and modular software systems using Python (FastAPI, Flask, Django REST Framework) and object-oriented design principles. Experienced in building scalable microservices, integrating relational and NoSQL databases, and automating data pipelines, with production deployment experience using Docker, Git/GitHub version control, and CI/CD (GitHub Actions). IBM SkillsBuild AI/ML Virtual Intern and Oracle Cloud Infrastructure Data Science Professional certified, with software delivered independently, in an internship setting, and through competitive hackathons.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript/TypeScript, SQL
Software Engineering: Object-Oriented Programming, REST API Design, Microservices, Modular System Design, Debugging & Troubleshooting, Version Control (Git/GitHub)
Backend & APIs: FastAPI, Flask, Django REST Framework, JWT Authentication, WebSockets, Swagger/OpenAPI Documentation
Databases: PostgreSQL, MySQL, MongoDB, Redis
Frontend: React, MERN Stack, HTML/CSS
DevOps & Cloud: Docker, GitHub Actions (CI/CD), AWS, Oracle Cloud Infrastructure (OCI)
Data & ML (applied): Pandas, NumPy, Scikit-learn, TensorFlow, Hugging Face, LangChain

PROFESSIONAL EXPERIENCE

IBM SkillsBuild (PBEL Program) – AI/ML Virtual Intern **Jun 2026 – Aug 2026**

Project : Recon – Cybersecurity Threat-Intelligence Backend System

- Designed and implemented a Python REST API (FastAPI) that serves a Random Forest classification model to detect 5 network-attack categories, with Swagger-documented endpoints for production consumption.
- Built modular, object-oriented data-preprocessing pipelines in Python (Pandas, Scikit-learn) implementing feature scaling, missing-value imputation, and correlation-based feature selection on high-dimensional telemetry data.
- Engineered a dual-server backend architecture – a FastAPI inference service plus a Flask dashboard – with pickle-based model versioning, delivering a working system within a 2-month Agile sprint.

ENGINEERING PROJECTS

S.A.F.E. – Real-Time Python Backend Pipeline (2026) – *Python, REST API, TensorFlow*

- Designed and deployed a real-time, end-to-end ingestion-to-inference backend pipeline in Python with automated model training, versioning, and a scalable REST API for serving predictions.

SentiScan – Flask REST API & NLP Engine (2025) – *Python, Flask, REST API*

- Designed and deployed a Flask-based REST microservice serving a Bidirectional LSTM sentiment-classification model, achieving 92.4% accuracy across 50K real-world reviews.
- Built and exposed real-time inference endpoints in Python for low-latency, production-style text-sentiment scoring.

MediPredict – Python ML Recommendation Engine (2025) – *Python, Scikit-learn, XGBoost*

- Engineered automated model-selection logic in Python across 7 medical prediction domains, achieving a 91.3% mean accuracy through ensemble logic.
- Built an interactive Python (Streamlit) interface delivering real-time, explainable predictions.

GPT LLM Playground – Transformer Built From Scratch (2026) – *Python, TensorFlow*

- Implemented a GPT-1-style transformer architecture from scratch in Python, building the attention and tokenization mechanics underlying modern generative-AI systems.
- Designed modular, reusable Python pipelines for data preprocessing, training, and inference, enabling reproducible experimentation.

EDUCATION

B.Tech in Computer Science – Anand Engineering College, Agra (AKTU) **2023 – 2027 (Expected)**

Intermediate (PCM) & High School (Science) – Shri Narsingh Bhagwan Inter College, Agra **2020 – 2022**

CERTIFICATIONS

- Oracle Cloud Infrastructure (OCI) Data Science Professional – Oracle (2025)
- Exploratory Data Analysis, Gold Certification (NSQF Level 5) – NASSCOM FutureSkills Prime, MeitY, Govt. of India (2026)
- SQL Certificate (Intermediate) – HackerRank (2024)

ACHIEVEMENTS

- Maintains 17+ public GitHub repositories spanning Python software projects, REST APIs, and applied machine-learning systems (github.com/HelloVishnu04).
- 2nd Place – Hack The Box TechTrix Cybersecurity Competition; Top 5 Finalist (Pan-India) – Virtual Lab Hackathon.
- Solved 200+ DSA problems across LeetCode & HackerRank, strengthening object-oriented problem solving, data structures, and algorithm design in Python.